# Fantasy Premier League Points Forecasting and Squad Optimization

Project status report

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## 1 Conclusion and status

The system - a per-position machine-learning ensemble feeding a MILP squad optimizer - beats its thesis-era LSTM predecessor by 29% in realized backtest points over an identical 31-gameweek window (1,966 vs 1,526; Section 5.1). The current state of play, most decision-relevant first:

1. **CatBoost (MAE loss) is now the production forecaster at every position**, selected empirically: a 20-gameweek walk-forward head-to-head and a static-window combination bake-off both found the single CatBoost model beats the NNLS blend, equal-weight top-k, and ridge stacking everywhere (weighted walk-forward MASE 0.684 vs the blend's 0.761) - the classic Clemen (1989) result that estimated combination weights lose to the best single member when members are many and collinear (Section 5.3).
2. **Best ranker, worst calibrated**: the new decision-aligned diagnostics show CatBoost ranks players and picks captains better than any blend (top-1 capture up to 0.57 vs 0.42) while under-predicting the aggregate point level by 40-55% - a direct consequence of its MAE loss fitting conditional medians of a zero-inflated target. Its level must be recalibrated before it feeds the MILP, whose transfer penalties are absolute-scale (Section 5.3, Section 6).
3. **The fixture-difficulty and minutes-projection features remain an unresolved trade-off**: they improved forecast accuracy at every position yet reduced realized backtest points 1,966 $\rightarrow$ 1,880 (Section 5.4). The mean-vs-median finding above is a plausible mechanism, and the diagnostics now exist to test it.
4. **No simple or econometric technique tested beats the ML models** (nine baselines, Section 5.2); a further 37 engineered features (opponent form, exponentially-weighted form, per-90 rates, xP) were roughly accuracy-neutral on the validation window - reported honestly, kept pending hyperparameter tuning, which remains unexplored territory.
5. A code-and-concept review found and fixed four methodological errors (blend-weight leakage, two live-mode staleness bugs, an evaluation asymmetry); the absolute backtest numbers predate the leakage fix and will be re-baselined (Section 6).

Priorities, in order: recalibrate CatBoost's level, then re-baseline the actual-points backtest under the fixed weight scheme with CatBoost-only forecasts; run the new Optuna tuner at scale; revisit the fixture/minutes divergence with the new diagnostics; then a per-player model-selection pilot.

## 2 Introduction

The system predicts FPL player points one to several gameweeks ahead and converts forecasts into weekly squad decisions (transfers, lineup, captaincy, chips) via a MILP following Kristiansen et al. (2018). The governing principle is honest empirical validation: every candidate technique is tested against the production system on a fixed backtest window, and negative results are reported rather than discarded.

# 3 Data

Per-gameweek player statistics come from Vaastav Anand's public mirror of the official FPL API (`vaastav/Fantasy-Premier-League`), spanning six seasons - 2020-21 through 2025-26, 162,981 player-gameweek rows. The two oldest seasons lack the Opta expected-goals family and `starts` (introduced 2022-23); these are treated as missing, which the tree-based models handle natively. Gameweeks carry a single ascending index (`GW_global`): dataset season $N$ occupies gameweeks $(N-1)\times 38+1$ through $N\times 38$, so the fixed validation window - 2024-25 season, gameweeks 1-31 - is GW153-183.

Exploratory analysis (`notebooks/eda.ipynb`, all six seasons) establishes the two facts the methodology is built on: `total_points` is heavily right-skewed and zero-inflated at every position (Shapiro-Wilk rejects normality, $p<0.001$ throughout), motivating a scale-free error metric and models free of Gaussian assumptions; and the distribution is stable across seasons, with one caveat - the average defender score-per-gameweek series fails an Augmented Dickey-Fuller stationarity test over the full six seasons ($p=0.41$; the other positions pass), plausibly reflecting the 2025-26 `defensive_contribution` scoring change.

# 4 Methodology

## 4.1 Feature engineering

Each player-gameweek row carries 115 engineered features:

- **Form.** Rolling 3- and 5-gameweek means, previous-gameweek values, and two expanding-mean horizons - season-to-date (resets each season) and career-to-date - over 18 per-gameweek statistics (points, minutes, xG family, BPS, ICT, price, ownership), plus exponentially-weighted means (halflife 3 gameweeks) for the ten most form-sensitive statistics, motivated by the earlier finding that exponential smoothing beats flat windows among simple baselines. All shifted one gameweek relative to the target row, so no row's features contain its own outcome.
- **Rates.** Per-90-minute rates (goals, assists, xG, xA over rolling minutes) separating production efficiency from playing time.
- **Opponent strength.** The upcoming opponent's rolling 6-gameweek attacking/defensive form and clean-sheet rate, computed at team level and merged by opponent with the same strict one-gameweek shift - a dynamic, in-season complement to the static FDR.
- **Fixture difficulty.** The official FPL Fixture Difficulty Rating (1-5) of that gameweek's opponent, plus the mean over this and the next two fixtures. Deliberately *not* shifted: fixture lists are published ahead of play, so these are known-ahead inputs, not leakage.
- **Minutes projection.** Rolling 5-game start rate and 60+-minute rate, shifted like form - a "will he actually play?" signal.
- **FPL's own forecast.** Shifted forms of xP, FPL's published pre-match expected points.

Statistics absent for entire early seasons (the Opta xG family and starts before 2022-23) are kept as missing values rather than zero-filled: "not collected" is a different fact from "recorded zero", and the gradient-boosted models branch on missingness natively.

## 4.2 Forecasting models

Four independent models, one per position (GK/DEF/MID/FWD) - the statistics that predict a goalkeeper's points are largely disjoint from a forward's, and pooling risks one position's scale dominating the loss. Per position, every regressor in a fixed registry is trained on the same features: LightGBM, XGBoost, CatBoost (MAE loss), OLS, Ridge, ElasticNet, PLS, Random Forest, Extra Trees, $k$-NN, LinearSVR, and a sample-capped RBF SVR. Boosted models take raw features including missing values; linear/kernel/distance models get a zero-imputer and standardizer. Registry members that tests find unhelpful are retained - the blend simply assigns them near-zero weight, preserving the comparison for later revisiting.

How the registry members are *combined* is itself an empirical question, answered per position by a combination bake-off scored on identical held-out rows: the best single member (no combination at all) vs. non-negative least squares (NNLS) vs. an equal-weight average of the top-$k$ members vs. ridge-regularised stacking - the latter two being the forecast-combination literature's standard remedies for the instability of estimated weights (Clemen, 1989). Whatever wins becomes the production combiner, so the choice tracks the evidence rather than an assumption. Weight fitting is separated by purpose: *evaluation* weights are fit on one half of the held-out test window and scored on the other half, so reported accuracy is a genuine holdout figure; *production and backtest* weights are fit on a 16-gameweek window strictly before whatever is subsequently predicted, so no weight ever sees a gameweek it will be used to forecast. Plain OLS is designated the *index* - the simple benchmark every technique must beat, as a passive market index is the bar an active strategy must clear.

Beyond MAE/MASE, evaluation includes decision-aligned diagnostics motivated by a structural trap: MAE-family metrics are minimized by conditional *medians*, while the downstream optimizer consumes conditional *means* and captaincy rewards the upside tail. The diagnostics - RMSE (mean-aligned), bias, total calibration (ratio of predicted to actual total points), within-gameweek Spearman rank correlation, and top-1 capture (how often the highest-ranked player actually scored most) - make that gap visible instead of letting MASE alone select median-flattening models.

A parallel probabilistic view - per-position LightGBM quantile regressors (p10/p50/p90, crossing repaired by row-wise sorting) - produces a prediction interval per player-gameweek, because two players with equal expected points are not equal decisions once the 2x captain multiplier rewards upside. It does not yet feed the optimizer.

## 4.3 Evaluation design

Three layers, in increasing decision-relevance:

1. **Static split.** Train on GW $\leq 152$, evaluate on GW153-183 - the window the original LSTM was validated on, keeping every comparison on one fixed window.
2. **Walk-forward validation.** For each gameweek from a start point, train strictly on earlier data and predict that gameweek only - many genuinely out-of-sample evaluations rather than one.
3. **Actual-points backtest (gold standard).** Walk-forward predictions run through the MILP over GW153-183; resulting squads scored with realized points. An accuracy gain that does not survive contact with the optimizer is not an improvement (Section 5.4).

Point forecasts are scored with MAE and, primarily, MASE (Hyndman & Koehler, 2006) - MAE relative to the in-sample error of a naive last-gameweek forecast, the scale fit on training data only; MASE < 1 beats the naive floor regardless of the target's scale. Probabilistic forecasts are scored with pinball loss and [p10, p90] coverage against its nominal 0.80.

### 4.4 Squad optimization

The MILP follows Kristiansen et al. (2018): 15-player squad (2/5/5/3) under budget, max 3 per club, 11-player lineup under formation constraints, captain/vice-captain, limited free transfers with a 4-point penalty per extra, and one-shot chip logic (two wildcards, free hit, bench boost, triple captain), solved as a rolling horizon with only the first gameweek's decision locked in. Venter & van Vuuren (2024) - whose formulation matches almost exactly - attribute their strong case study (top 4.08% of  8.24M managers) to lookahead and forecast quality, not optimizer sophistication; hence this project's focus on forecasting.

## 5 Results

### 5.1 Backtest lineage

All systems scored by realized points over 2024-25 GW1-31, identical optimizer:

| System | Actual points |
|---|---:|
| Old LSTM + MILP (thesis) | 1526 |
| LightGBM + MILP, 4-season history | 1811 |
| Ensemble + MILP, 4-season history | 1900 |
| Ensemble + MILP, 6-season history | **1966** |
| 1. fixture/minutes features | 1880[1] |

Extending history from four to six seasons was an unambiguous win: MASE fell at every position (FWD below 1.0 for the first time) and realized points rose 1,900 → 1,966.

### 5.2 Forecasting-technique experiments

Eight simple and econometric baselines - rolling mean, naive drift, SES, Holt, Theta, Croston, pooled AR(1), per-player ARIMA(1,0,1), and an empirical-Bayes shrinkage of player means toward position means - were each tested per position. None beats the ML models anywhere; SES is the best simple method, Croston and EB-shrinkage the weakest (full tables in `RESEARCH_LOG.md`). Plain OLS, the index, beats every simple baseline; the ML models in turn beat OLS.

### 5.3 Expanded model registry

Six techniques were added in one round (LinearSVR, RBF SVR, XGBoost, CatBoost, PLS, EB-shrinkage) and evaluated on the standard static split:

| Position | CatBoost | LinearSVR | 12-member ensemble | previous ensemble |
|---|---:|---:|---:|---:|
| GK | **0.513** | 0.513 | 0.534 | 0.588 |
| DEF | **0.705** | 0.713 | 0.747 | 0.799 |
| MID | **0.731** | 0.751 | 0.806 | 0.799 |
| FWD | **0.849** | 0.869 | 0.853 | 0.959 |

*MASE, GW153-183 static split. CatBoost uses MAE loss; ensemble scored on a genuine holdout half-window.*

[1] Unresolved regression despite improved forecast accuracy (Section 5.4). Both this and the 1,966 figure predate the blend-weight fix of Section 6 and are modestly inflated; their relative comparison is unaffected.

Two findings. First, **CatBoost with MAE loss is the best single model at every position**, by a wide margin over everything preceding it - consistent with the hypothesis (raised by LinearSVR's strong showing) that MAE-aligned training objectives suit a zero-inflated target with outlier hauls better than squared error. PLS and XGBoost added nothing (retained as near-zero-weight members). Second, **the blended ensemble now trails standalone CatBoost everywhere** - with 12 collinear members, NNLS overfits its half-window weight fit (the MID blend failed to select CatBoost at all).

That second finding was settled by two independent tests. A 20-gameweek walk-forward head-to-head (GW169-226, members retrained each step, weights fit strictly before the window) had CatBoost-only beat the blend at every position - weighted MASE 0.684 vs 0.761 - and the combination bake-off on the static window agreed, with equal-weight top-$k$ second and ridge stacking last:

| Position | CatBoost only | top-k (k=3) | NNLS | ridge |
|---|---|---|---|---|
| GK | **0.502** | 0.517 | 0.547 | 0.551 |
| DEF | **0.689** | 0.705 | 0.782 | 0.801 |
| MID | **0.702** | 0.720 | 0.797 | 0.780 |
| FWD | **0.788** | 0.805 | 0.884 | 0.887 |

*Combination bake-off: MASE on identical held-out rows (GW168-183 eval half). The best single member beats every combination scheme - Clemen's (1989) forecast-combination result in the wild.*

Production forecasting is therefore CatBoost-per-position, chosen by the bake-off on every training run rather than hard-coded - if a future change makes a combination win again, production follows automatically.

The decision-aligned diagnostics add a critical nuance: **CatBoost is the best ranker but the worst calibrated**. It captures the actual top scorer far better than the blend (top-1 capture 0.57 vs 0.42 at MID; higher at every position) with near-equal rank correlation - yet its bias is $-0.32$ to $-0.60$ points per player-gameweek and its forecasts sum to only 44-63% of the points actually scored, exactly the median-flattening the MAE loss predicts on a zero-inflated target. Ranking quality is what transfers and captaincy consume, so this is the right model to keep - but the MILP also makes *absolute-scale* decisions (a 4-point transfer penalty, chip thresholds), which deflated forecasts would distort. A level recalibration (scalar or isotonic, fit on the same pre-window holdout as the weights) is required before the backtest re-baseline (Section 6).

A further honest null result: the 37 new features added alongside this batch (opponent form, EWMA form, per-90 rates, xP) left CatBoost's full-window MASE essentially unchanged (0.513→0.517 GK, 0.705→0.706 DEF, 0.731→0.730 MID, 0.849→0.847 FWD). They are retained - the underlying semantic corrections are correctness fixes regardless, and no hyperparameter tuning has yet been attempted that might exploit them - but they have not yet earned their keep on accuracy alone.

## 5.4 Fixture/minutes features: better forecasts, worse squads

Adding fixture-difficulty and minutes-projection features improved MASE at every position (DEF most, consistent with clean-sheet dependence) - yet realized backtest points *fell* 1,966 → 1,880. Leading hypothesis: the features smooth the mean forecast toward safe, nailed players, and the MILP - blind to variance - loses the high-ceiling captaincy differentials that drive realized hauls. The features are committed but not declared a net win; resolving this (noisecheck on a second window, probabilistic-upside captaincy) is a top open item. The probabilistic module's [p10, p90] coverage is 0.88-0.93 against nominal 0.80 - somewhat wide, usable for relative risk ranking; conformal calibration is a possible refinement.

### 5.5 Exploratory branches

Two isolated branches, neither merged: a Bayesian belief-state MDP manager after Matthews et al. (2012) - 1,083 points myopic / 847 Q-learning where the production system scored 1,900, expected given documented simplifications - and a per-player model-selection plan (feasibility capped by 35% of live players having no prior-season history; recommendation: single-position pilot).

## 6 Methodological limitations

An internal code-and-concept review (2026-07-04) found four errors; all four were fixed the same day. One further limitation emerged from the diagnostics added afterwards. Recorded here because published numbers predate some fixes:

1. **Production forecast level is miscalibrated (open, fix designed).** The production CatBoost models under-predict the aggregate point level by 40-55% (a structural consequence of MAE loss on a zero-inflated target), while ranking players better than any alternative. Harmless for pure ranking, but the MILP's transfer penalty and chip logic operate on the absolute point scale, so a level recalibration on the pre-window holdout must precede the backtest re-baseline.
2. **Blend-weight leakage (fixed).** Backtest predictions formerly reused blend weights fit inside the backtest window itself. Weights are now always fit on a window strictly before whatever is predicted. The 1,966/1,880 figures predate the fix and are modestly inflated; their relative comparison stands (both leaked identically). Re-baselining is pending.
3. **Live-mode staleness (fixed).** The weekly driver formerly reused each player's last played row, so live fixture-difficulty described last week's fixture and rolling form excluded the most recent match; API-vs-dataset team-name mismatches also silently dropped some teams' players. Live predictions now use per-gameweek API fixture difficulties and a synthetic next-gameweek row whose features include everything played. Backtests were never affected.
4. **Index-check asymmetry (fixed).** The ensemble-vs-OLS verdict now compares both on the same held-out rows.
5. **Optimizer rule currency (open).** The MILP caps banked free transfers at 2 (now 5 in FPL) and assumes sales at market value (real FPL: purchase price plus half profit). Fine for historical comparison; wrong for live 2026-27 play. Deliberately deferred - optimizer work is deprioritized in favor of forecasting.

# 7 References

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